Impact of Large Language Models on UCSD Student Performance

GROUP 8

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Questions we will cover

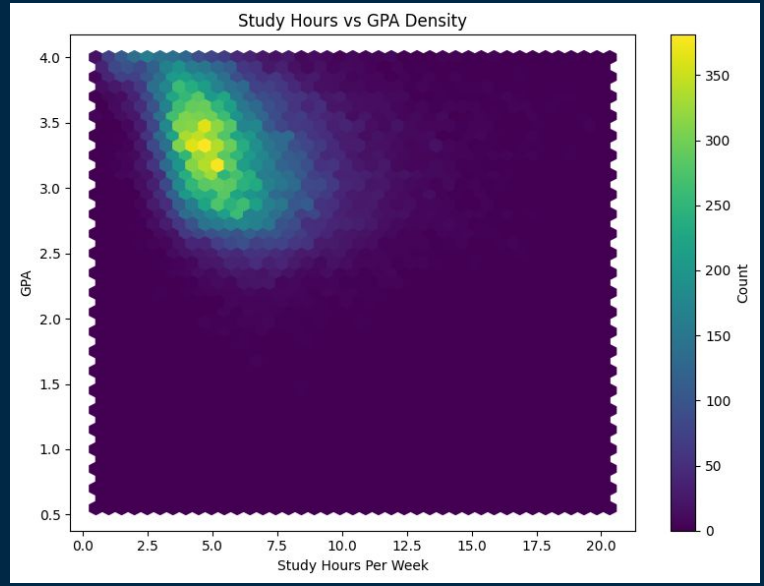
Main Questions

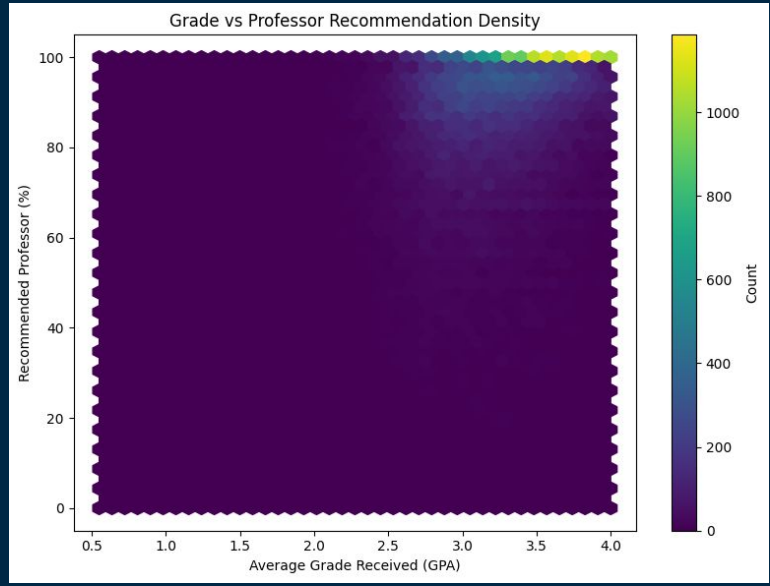
- How do perceived difficulty and quality of course change with each LLM release?
- How does average received grades change with LLM releases?

Sub questions

- How do students rate professors based on their received grade?
- What does the average GPA look like across departments?
- How did COVID affect average GPAs?
- Cross reference COVID vs LLM effect on difficulty

Plots Analysis – CAPEs





Dataset Analyzed from CAPEs:
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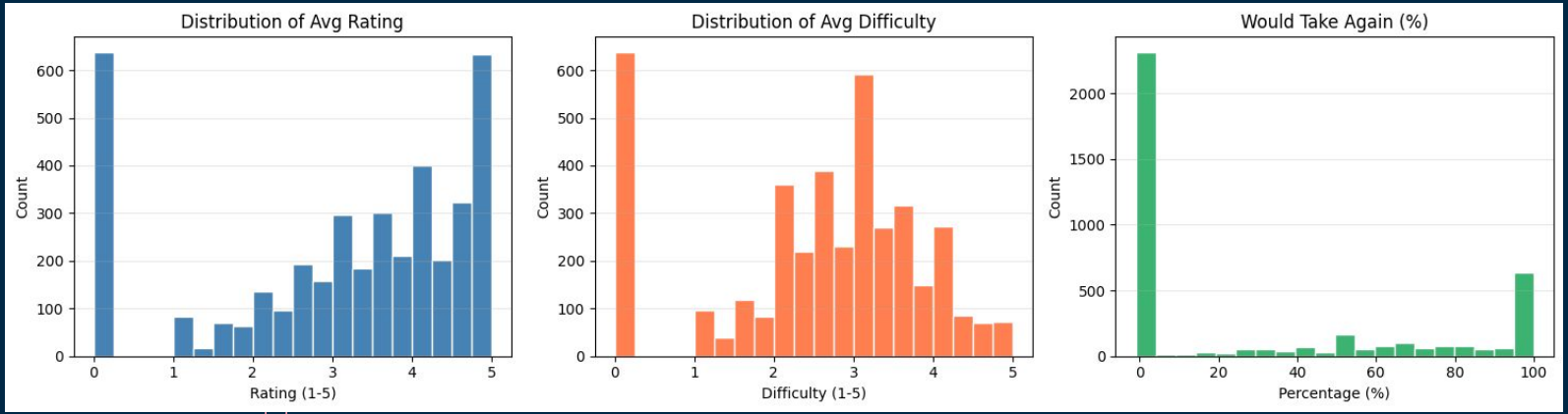
Plots Analysis – RateMyProfessors

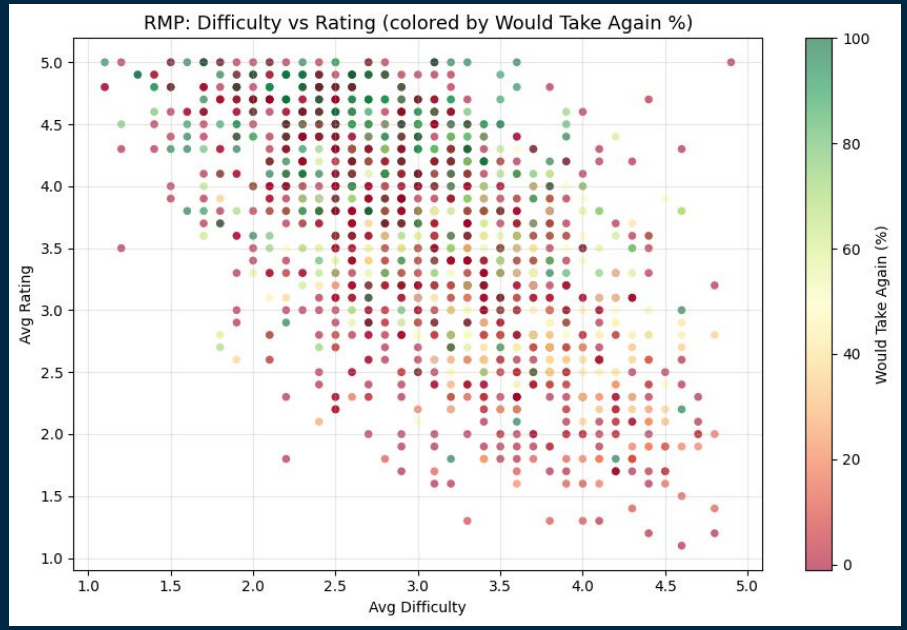
- Due to online and anonymous options, the surveys are rather extreme
- Plots show that most ratings are either lowest or highest



Dataset Analyzed from RMP:
https://github.com/mmerioles/scrape-rmp
(made by Matthew)

Plots Analysis – RateMyProfessors

- Higher rating of professors leans towards lower difficulty of courses
- Potential bias of surveys



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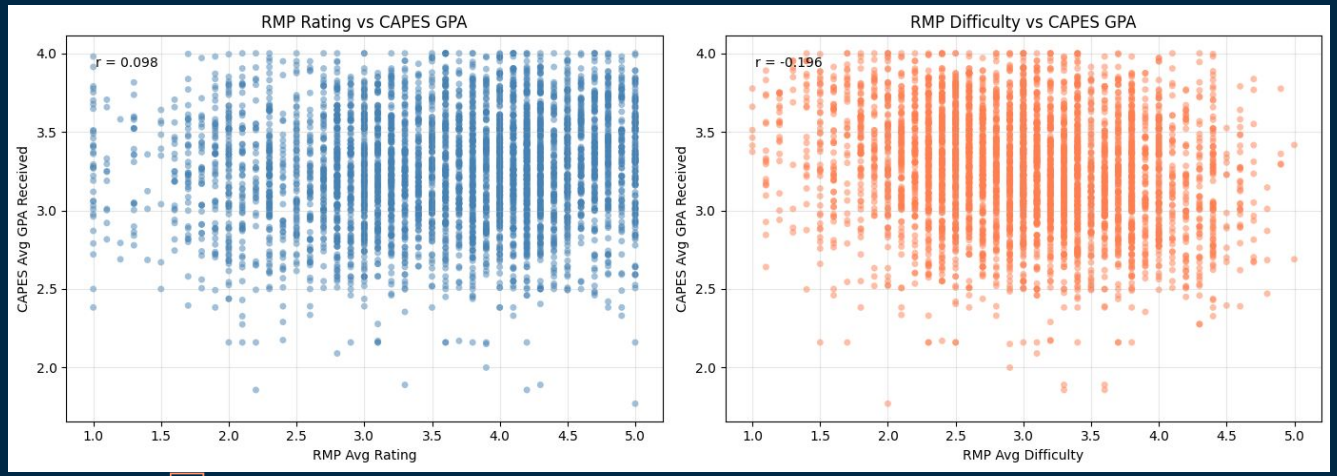
Plots Analysis – Cross Reference

- Comparison plots show that RMP's average rating is proportional to CAPEs' average GPA received
- Also, RMP's average difficulty is inversely proportional to CAPEs' average GPA received



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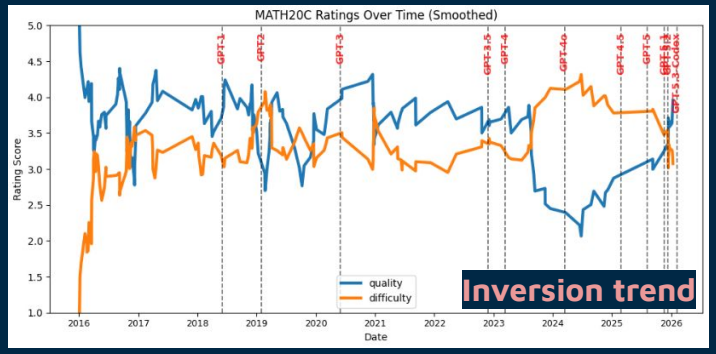
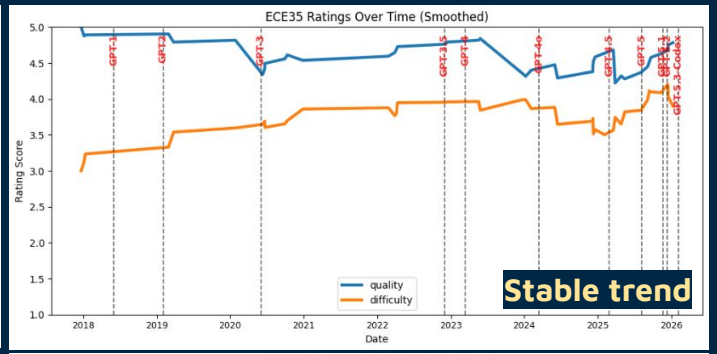
How do perceived difficulty and quality of courses change with each LLM release?

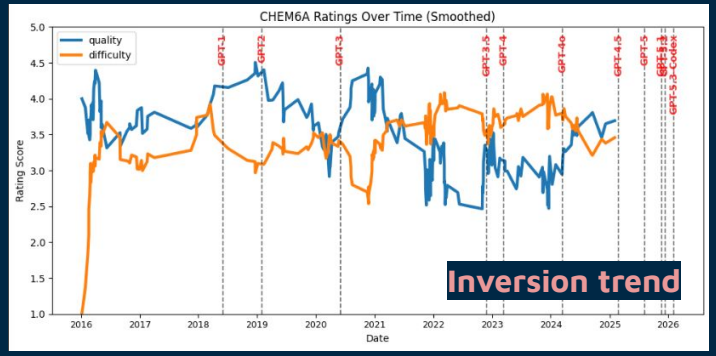
- Hypothesis: Courses should be perceived as easier because people have access to these new "study tools"... Course quality should also improve as well... right?

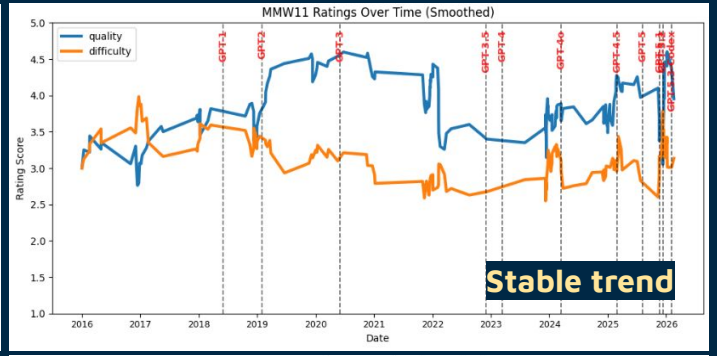
Grok

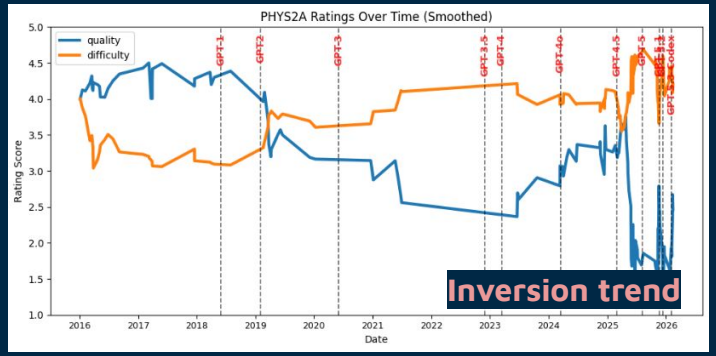
Gemini

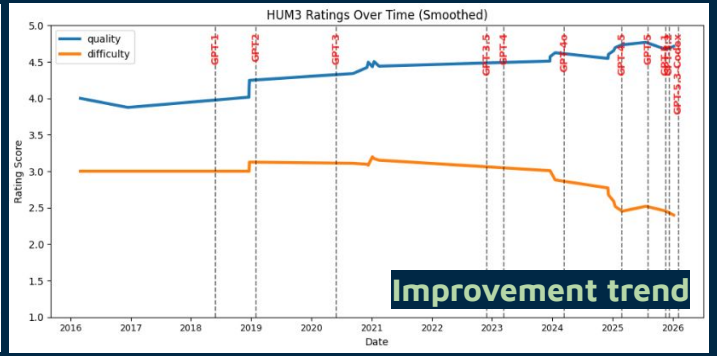








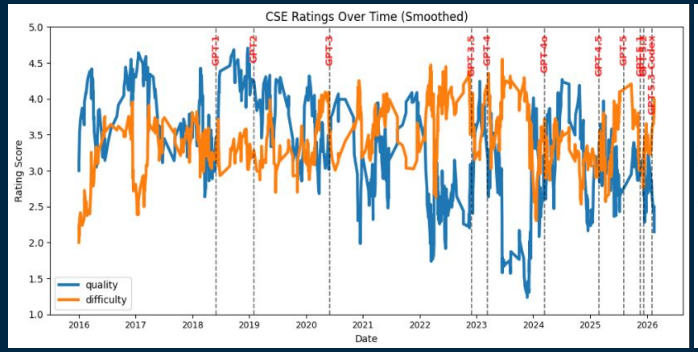


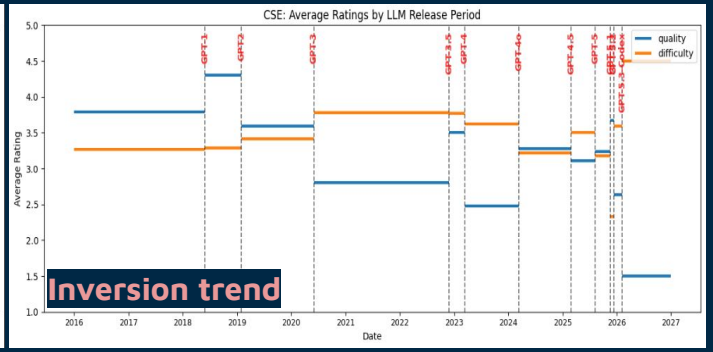


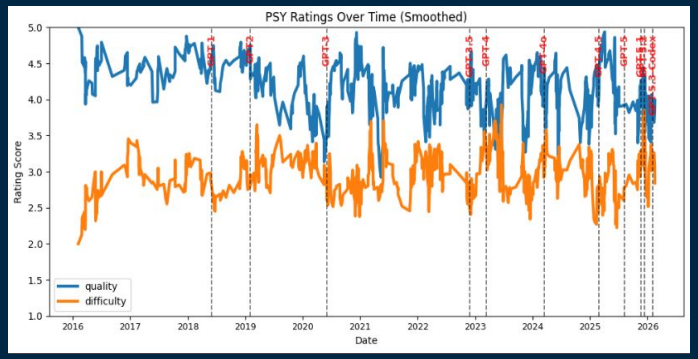
Select Class Behavior

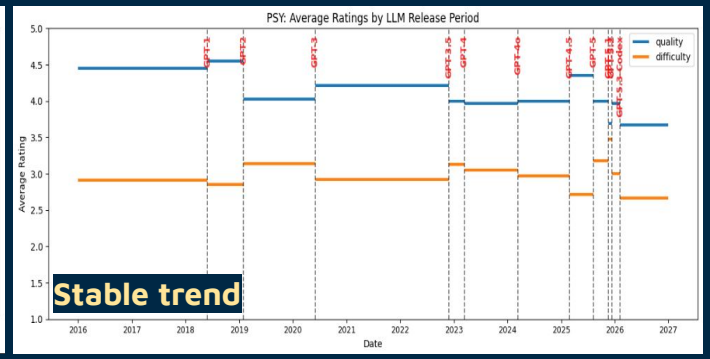
- **Inversion trends**: Quality perception rising while difficulty increases
- **Stable trends**: Quality and difficulty remain relatively unchanged
- **Improvement trends**: Difficulty decreases while perceived quality increases

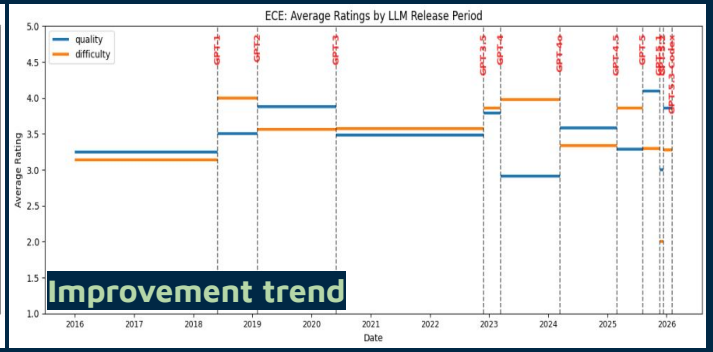
Question: How does this look like across the departments?











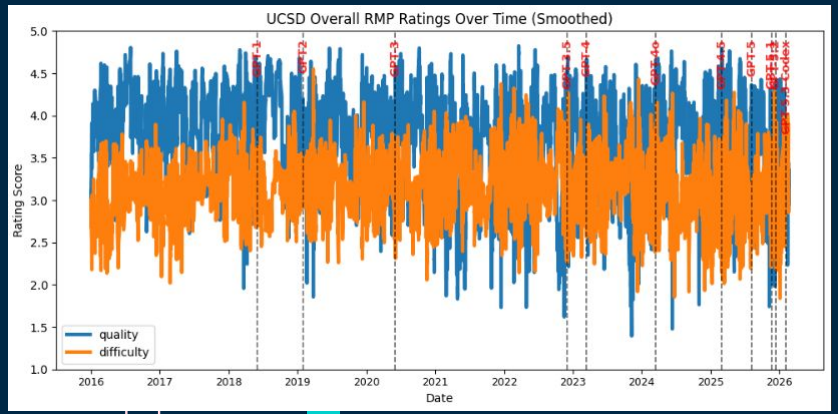
Department Averaging

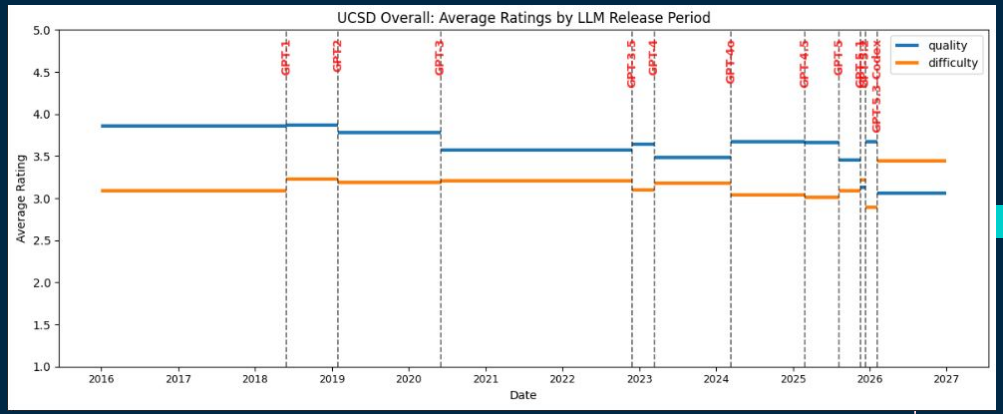
- Accumulating reviews over departments shows similar cases of **Inversion**, **Stable**, and **Improvement** trends.
- But compiled, we see
  - 21 Inverting
  - 40 Stable
  - 25 Improving
- RMP data reveals consistent department-level trends in perceived quality and difficulty.

How do perceived difficulty and quality of courses change with each LLM release?

Overall, RMP rating remain relatively stable over time!

- Student perception of teaching doesn't change dramatically
- External factors may not strongly affect ratings
- Aggregation smooths out course-level variation





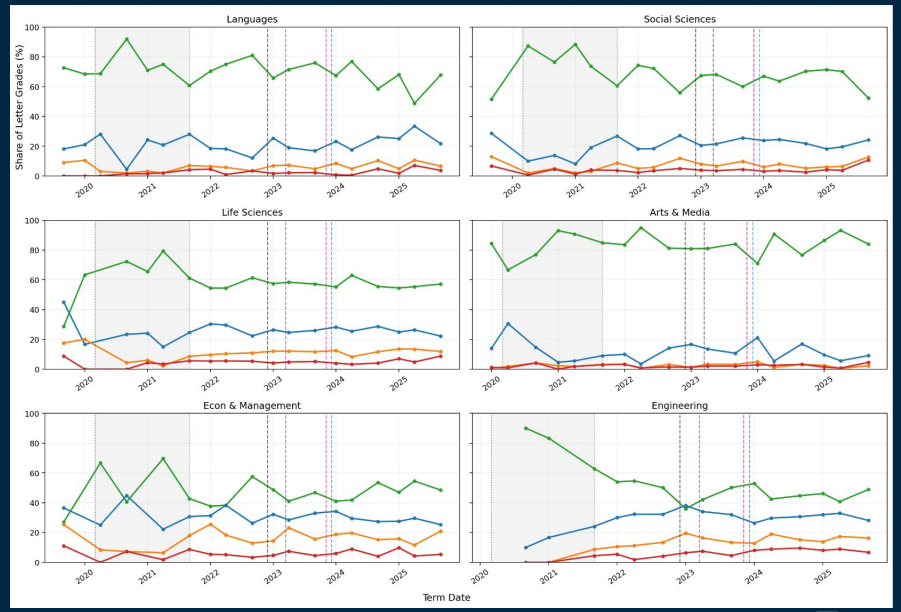
Average GPA Changes w/ LLM Trend & COVID

- LLM started rising in early COVID years
- Most, if not all departments, have visibly increasing average GPA across first few years of COVID
- This trend starts going the other way after few quarters of 2020

Dataset Analyzed from SunSET:
https://github.com/SheepTester/ucsd-sunset

Average GPA Changes w/ LLM Trend & COVID

- This demonstrates how both COVID and LLM affected grades and class structures
- After realizing, professors and departments took actions making stricter rules, lowering the GPA



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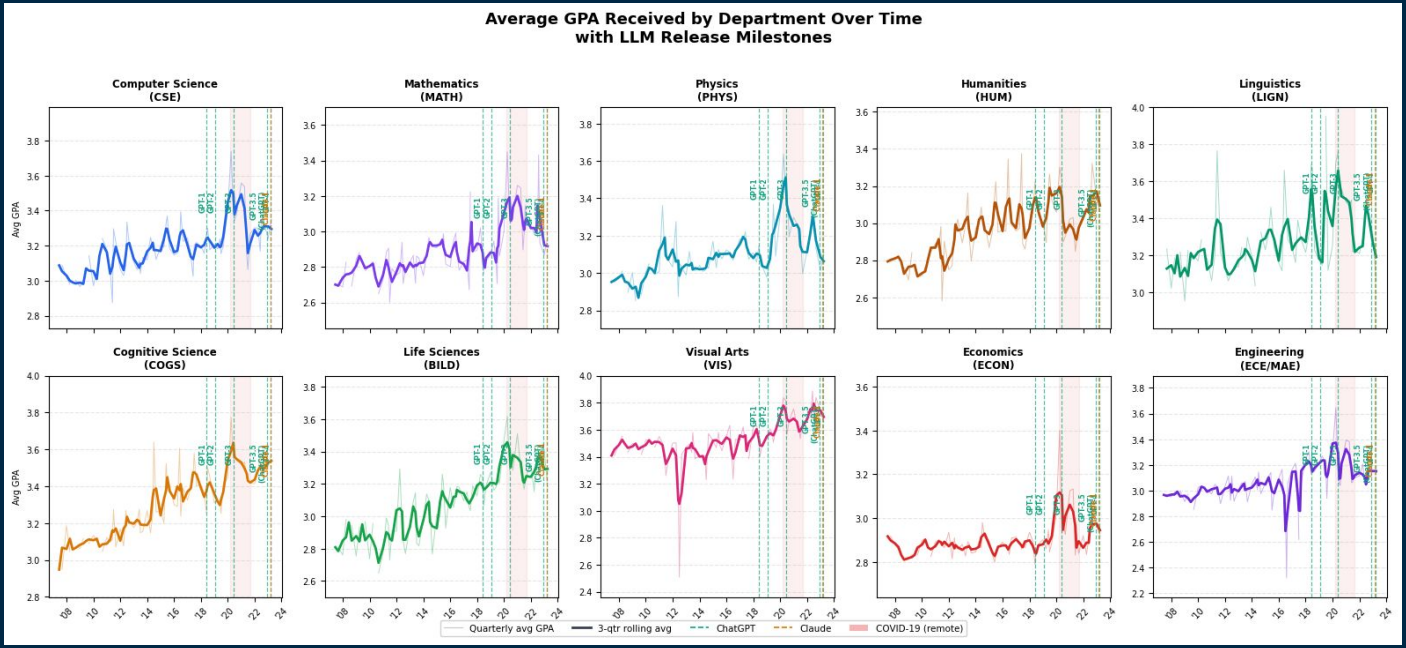
Additional Plots Regarding LLM Releases



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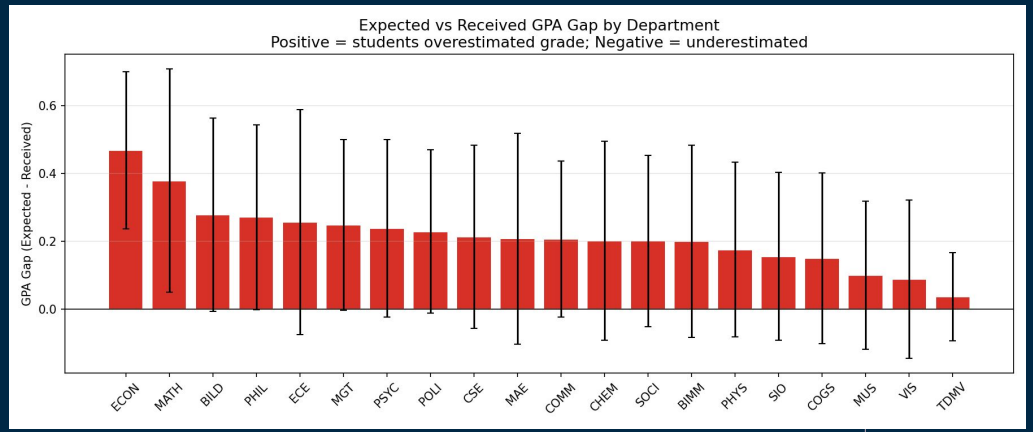
Students in Harder Departments Overestimate Their Grades the Most

- Every department shows a positive gap — students universally overestimate their grades before results are released
- ECON and MATH students overestimate by ~0.47 and ~0.38 GPA points respectively — the largest gaps across all departments
- Arts departments (TDMV, VIS, MUS) show near-zero gaps, suggesting grades align closely with student expectations
- Higher gap = harder grading environment, making these departments more likely to seek AI assistance



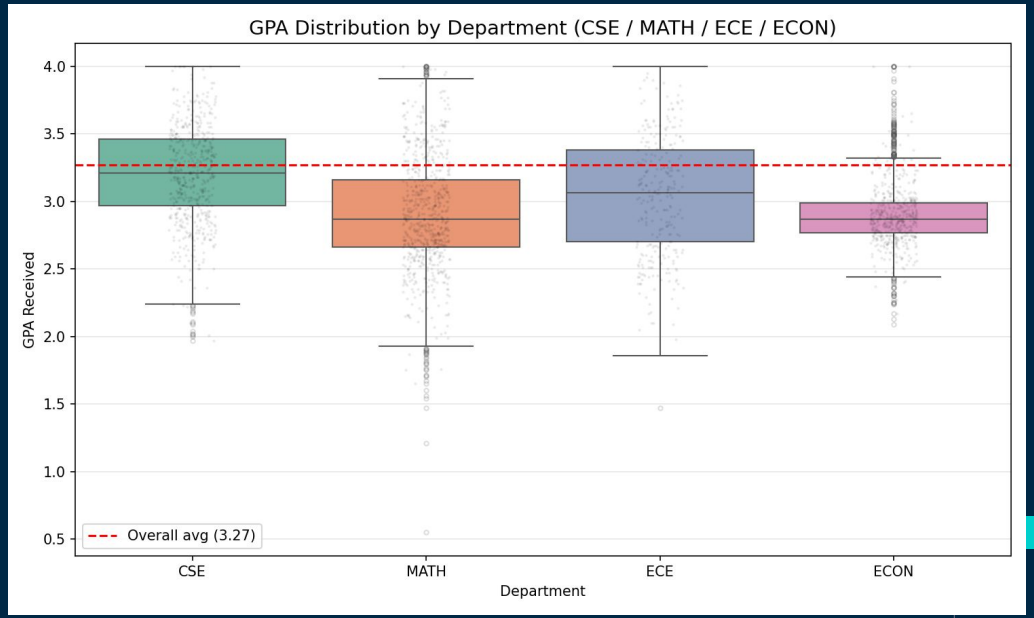
Dataset Analyzed from UCSD CAPEs:
https://cape.ucsd.edu

STEM Departments Show Lower & More Volatile GPA Distributions

- CSE scores closest to the UCSD average (3.27),
  but with a wide range from 2.25 to 4.0

- MATH and ECON have the lowest medians (~2.85),
  with MATH showing the largest spread — indicating high grading inconsistency

- All four STEM departments fall below the overall average, confirming a structural grading gap vs. other disciplines



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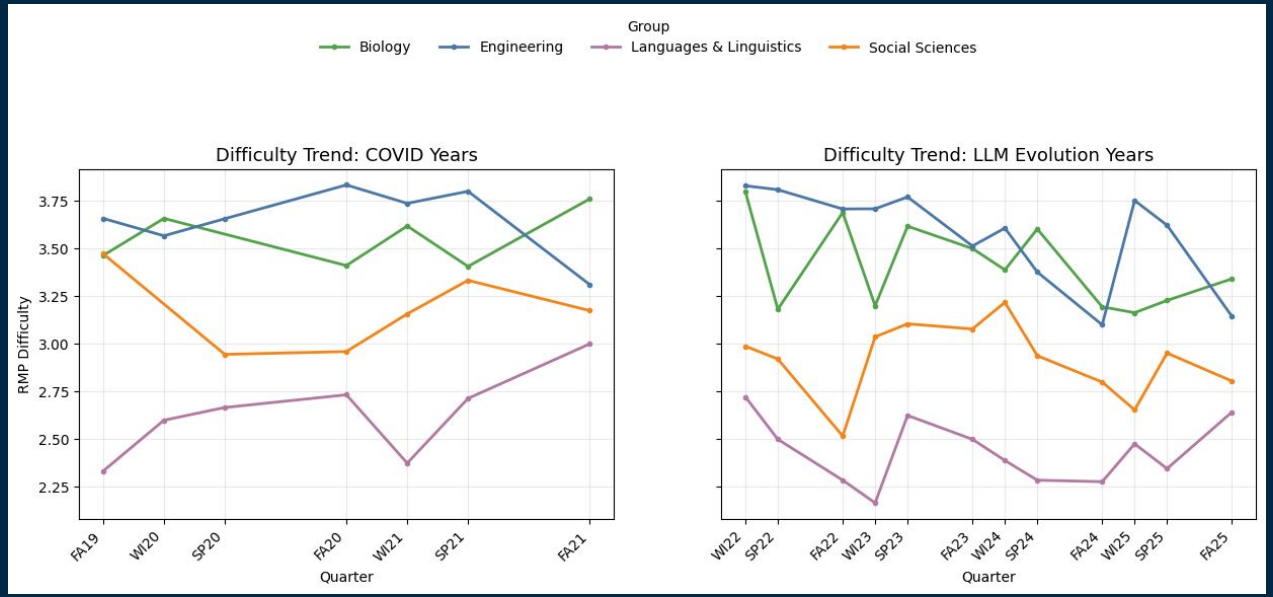
Cross reference COVID vs LLM effect on difficulty

Hypothesis: The COVID period had a stronger effect on perceived course difficulty than the later LLM-evolution period, and this effect differed by broad subject group.



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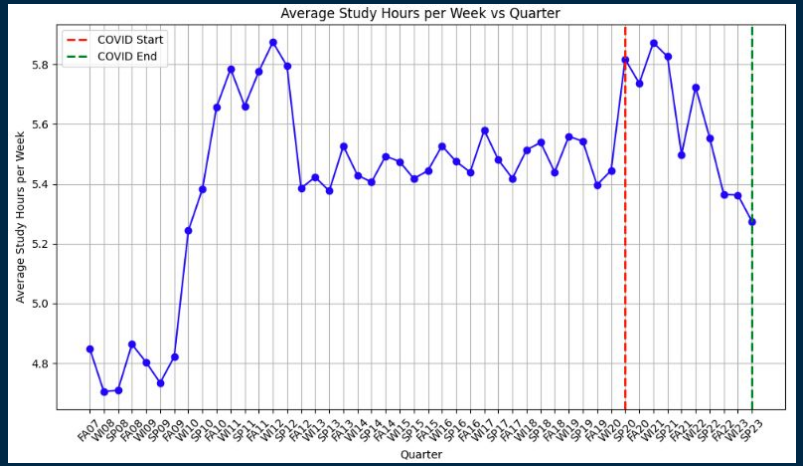
COVID Impact On Average Study Hours Per Week



Dataset Analyzed from CAPEs: https://www.kaggle.com/datasets/sanbornpnguyen/ucsdcapes/data

- The average study hours per week shows a sharp rise during the start of COVID post which it declines towards the end of COVID period.

Key Takeaways

1. **LLM releases did not produce a universal shift in course perception.** Across departments, trends were mixed: some courses looked easier, some stayed stable, and others showed inversions between quality and difficulty.
2. **COVID appears to have had the stronger immediate impact on grades.** Average GPA rose noticeably during the early pandemic, then began to normalize as departments adjusted grading and course policies.
3. **Department context matters.** Harder STEM-heavy departments showed lower and more volatile GPAs, and students in some departments overestimated their final grades the most.
4. **RMP and CAPEs tell a consistent story.** Higher professor ratings generally aligned with higher received GPA and lower perceived difficulty, though survey bias likely affects extremes.
5. **Overall conclusion:** The effect of LLMs on student outcomes is not simple or uniform; institutional changes, department differences, and the COVID period all shape the observed trends.

Thank you!